Lesson 3-5 • Day 3 • Apply / Performance Task

Observation Lesson • 65 min • Mirrors admin's Lesson Plan Template (Teachers will / Students will / Questions to consider)

Phase plan (65 min): 7:45–7:55 Do Now (10) • 7:55–8:00 Launch (5) • 8:00–8:15 Fireworks (15) • 8:15–8:35 Venetta Performance Task (20) • 8:35–8:45 Share/Summary (10) • 8:45–8:50 Exit (5)

Math Obj: (1) Identify zeros via factoring or synthetic division. (2) Use zeros to graph a polynomial. (3) Interpret features in a real-world context.
Language Obj: Explain solutions verbally and in writing using sentence frames + structured talk-time.
Essential Q: How are the zeros of a polynomial related to its equation and graph?

1. Do Now — SAT practice from 3-5 — 7:45–7:55 • DOK 2 • 10 min

Less than 10 min • low-floor entry

Item: SAT practice (3-5)
Without using a graphing calculator, determine which of the given graphs matches $f(x) = x^3 + x^2 - 4x$. Explain why you made that choice.

[SPEAK] Teachers will...

- Display problem on board
- Read prompt
- Tell students they have 3-5 minutes to work on problem (*no calculator)
- Set timer for 3 minutes independently think/write
- After timer, 3 minutes to pair up and share answers with each other
- After 5 minutes, class discussion

[STUDENT] Students will...

- Actively listen to prompt
- Record initial ideas on their paper (3 min)
- Share with a partner and record explanation for their choice (3 min)
- Share out in class discussion (2-3)

[PUZZLE] Questions to consider...

- How can you engage students in the lesson?
- How can you hook them?
- How are all adults intentionally planned for?

[STAR] Answer key

Factor: $f(x) = x(x^2 + x - 4)$. Zero at $x = 0$ from the linear factor; quadratic factor has discriminant $1 - 4(1)(-4) = 17 > 0$, giving $x \approx -2.56$ and $x \approx 1.56$. Three real zeros total. Degree 3 + positive leading coefficient ⇒ end behavior DOWN–UP. Eliminate graphs that miss $x = 0$, show wrong end behavior, or have fewer than 3 zeros.

2. Launch — Unpack objectives + connect to prior — 7:55–8:00 • DOK 1-2 • 5 min

5–15 min • academic conversation

[SPEAK] Teachers will. . .

- Teacher asks for volunteers to read objectives
- Connect to previously learned skills / why we're doing what we're doing

[STUDENT] Students will. . .

- Volunteers read objectives

[PUZZLE] Questions to consider. . .

- What will you explicitly model and how?
- How will you know if students are ready to work alone?
- How are all adults intentionally planned for?

3. Explore — Fireworks Problem 8:00–8:15 • DOK 2 • 15 min

15 min • TWPS • cognitively demanding

Item: Fireworks: $h(t) = -16t^2 + 32t + 6.5$
Three parts: (a) reasonable domain, (b) zeros + meaning, (c) vertex + meaning. Apply / Make Sense and Persevere section of the student packet.

[SPEAK] Teachers will. . .

- Teacher asks student to read the problem(s). Cue up students to TWPS.
- **Questions to ask:** Discuss domain. Why is it important to think about the domain before starting? Why would only values > 0 work? What is domain talking about in this case? What are possible values for t? Can we have negative time? Why/why not?
- Instruct students 3-4 minutes to think/pair and prepare to share with class
- Circulate while students are thinking/pairing to identify students to warm-call to share out
- Ask students to share out
- Display on Desmos
- Release students to work on B and C in pairs. Communicate clear expectations (5-min timer to let them productively struggle). Add 2-3 more minutes as needed
- Circulate to ask open questions, assess progress, redirect as needed, clarify common misconceptions as a class
- Ask students to share their thinking (whoever did not participate in part A)

[STUDENT] Students will. . .

- Actively listen to problem being read
- Respond to questions the teacher asks
- Think independently (1 min)
- Pair up w/partner and share thinking (2 min)
- Students share their thinking when called on (2-3 min)

[STAR] Answer key

(a) Domain: $0 \le t \le\approx 2.19$ s (launch to ground impact). $t < 0$ has no physical meaning.
(b) Zeros: Solve $-16t^2 + 32t + 6.5 = 0 \Rightarrow t \approx 2.19$ s (the negative root ≈ -0.19 s is rejected). The positive zero is the moment the firework hits the ground.

(c) Vertex: $t = -b/(2a) = -32/(-32) = 1$ s. $h(1) = -16 + 32 + 6.5 = 22.5$ m. Vertex represents **maximum height** (22.5 m at $t = 1$ s).

4. Explore — Performance Task: Venetta Deli [STAR] DOK-3 DRIVER 8:15–8:35 • DOK 3 • 20 min

*15–20 min • TPS • DOK-3 driver • *Connect back to PBL*

Item: Venetta profit: $P(t) = t(t+3)(t-2)$ **[STAR] DOK-3**
Profit in hundreds of dollars, t years after 2000. **(A)** Sketch + confirm with Desmos. **(B)** During what years did Venetta NOT make a profit? **(C)** Predict 2020 profit if the model is appropriate.

[SPEAK] Teachers will...

- **Connect it to real-world:** "Does anyone know anyone that owns or operates any food businesses/restaurants/shops?"
- "Well, it can be stressful—business owners aren't always profitable. Why could that be? This is what this could look like in real life—but with Algebra."
- "Does anyone remember the profit equation from PBL/project?" *Connect back to PBL
- Teacher frames the problem (3-4 min): here's another profit equation for another store (not pizza, but deli sandwiches this time)
- What variables do we see here? What does t mean?
- Read task itself and then go to part A (5 min). **Question:** How can you identify the zeros from the factored form? How can you determine the behavior?
- **Summarize:** show student work from iPad and ask students "what do you think about this?" "do you agree or disagree?"
- B/C—Read and release students to think/pair/share their thinking (∼7 min)
- **Question:** How do you know, looking at a graph, if you have a profit?
- For part C, explore a variety of pathways to get that answer

[STUDENT] Students will...

- Engage in discussion around food businesses/profits/etc.
- Recall prior knowledge to PBL project
- Work in pairs/triads on B and C
- Engage in discussion around B and C with one another, and respond to questions from teacher as needed
- Share their thinking with the class

[PUZZLE] Questions to consider...

- What will students do for most of the session? What will they practice? How?
- What type of academic conversations will students engage in?
- How will you be interacting with them?

[STAR] Answer key

(A) Zeros: $t = 0, -3, 2$. Only $t \geq 0$ meaningful. Graph dips below t-axis on $0 < t < 2$, crosses up at $t = 2$, climbs as a cubic with + leading coefficient.
(B) No profit: on $0 < t < 2 \Rightarrow$ **years 2000–2001** (broke even at start of 2002).
(C) 2020 prediction: $t = 20 \Rightarrow P(20) = 20 \cdot 23 \cdot 18 = 8280$ hundreds of dollars = \$828,000. **CAVEAT (DOK-3 pivot):** 20-year extrapolation overshoots dramatically—cubic growth, market saturation, competition. *Discuss model limits.*

[WARN] Warning / Variant

Part C is the DOK-3 pivot — students must JUDGE model validity, not just compute $P(20)$. Press the model-limits discussion or this stays at DOK-2.

5. Share / Summary 8:35–8:45 • DOK 2 • 10 min

5–15 min • synthesis • return to objectives

[SPEAK] Teachers will. . .

- After ∼7 min, call group back together to report their thinking/findings. **Ensure equity of voice.**
- Share key takeaways

[STUDENT] Students will. . .

- Report thinking and findings to class
- Engage in equity of voice — listen and let others share

[PUZZLE] Questions to consider. . .

- How will students express what they have learned? This should be directly related to the mini-lesson and objective.

[CHAIN] Bridge

Bridge to Mon 5/11 (P3-cont, Storage Box #27): same factored-form reasoning, but now zeros are physical dimensions of a box.

6. Exit Ticket — Baseball + reflection 8:45–8:50 • DOK 2 • 5 min

5 min • DOK-2 cool-down

Item: Baseball — explain why graph shows only one zero
Height of a baseball thrown in the air, modeled by a quadratic. **Explain why the graph shows only ONE zero.**
Hint: domain restriction.

[SPEAK] Teachers will. . .

- Project exit prompt. Students complete independently and turn in at the door
- Tonight: scan answers; if many miss the domain-restriction idea, open Mon 5/11 with a Do Now bridge

[STUDENT] Students will. . .

- Complete the Baseball reflection independently
- Submit at the door

[PUZZLE] Questions to consider. . .

- How will students show what they've learned based on the objective?

[STAR] Answer key

A quadratic can have two mathematical zeros, but only one is in the meaningful domain ($t \geq 0$). The other zero is at **negative time** — before the ball was thrown — which has no physical meaning. The graph shows only the zero where the ball lands. Same idea as Fireworks (a) and Venetta (B): real-world contexts restrict which mathematical zeros are physically relevant.